



Motilal Nehru National Institute of Technology Allahabad

Department of Computer Science and Engineering

Introduction to Artificial Intelligence and Machine Learning (CSN12601)

Lab Manual: Assignment-1

Lab Title: Programs on Data Types, Arithmetic Operations, and Operators in Python

Objective:

1. To familiarize with the installation of Python and understand different ways of running Python programs.
2. To demonstrate different data types and type conversion functions in Python.
3. To understand and implement arithmetic and logical operations in Python.
4. To explore the use of bitwise operators in Python.
5. To practice string manipulation in Python.

Materials Required:

- Computer with internet access
- Python (latest version)
- Python Integrated Development Environment (IDLE)
- Access to an online Python environment (e.g., Google Colaboratory)

Theory:

Python is a high-level, interpreted programming language known for its simplicity and versatility. It supports multiple data types, including integers, floats, complex numbers, and strings. Understanding data types, type conversions, arithmetic operations, logical operations, bitwise operations, and string manipulations are fundamental to programming in Python.

List of Programs in Assignment-1

- 1) Installation of Python and demonstration to run programming using Interpreter, IDLE, Command Prompt, and Google Colaboratory.
- 2) Write a program to demonstrate different number datatypes in python.
- 3) Python Programs to demonstrate of various Type Conversion Functions.
- 4) Write a program to perform different arithmetic operations on numbers in python.
- 5) Write a program that include operators like “and”, “or”, “not” and Bitwise operators
 - a) Bitwise and (&)
 - b) Bitwise or (|)
 - c) Bitwise ex-or (^)
 - d) Bitwise complement (~)
 - e) Bitwise left shift Operator (<<), and Bitwise right shift Operator (>>).
- 6) Write a program to create, concatenate and print a string and accessing substring from a given string.